Optimizing User, Group, and Role Management with Access Control and Workflows

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled **Optimizing User, Group, and Role Management with Access Control and Workflows** is the Bonafide work of **Adchaya J (960222243006), Afrah Riyas (960222243007), and Akishma S (960222243008)** who carried out the work under my supervision.

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Submitted for the Naan Mudhalvan NM1051 – Service Now Administrator Project Work held at Arunachala College of Engineering for Women on ___________________.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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Abstract

The project titled **“Optimizing User, Group, and Role Management through Access Control and Workflows”** aims to design and implement a structured and secure access management system using the **ServiceNow** platform. In most organizational environments, unclear role assignments and poor access control mechanisms lead to inefficiencies, miscommunication, and data security issues. To overcome these challenges, this project focuses on establishing a systematic model that clearly defines users, groups, and roles, supported by automated workflows and Access Control Lists (ACLs).

The system is built around a simple project scenario involving a **Project Manager (Alice)** and a **Team Member (Bob)**. Each user is assigned specific roles and permissions to ensure that access to data and system functionalities aligns with their responsibilities. The **Project Manager** can create, assign, and approve tasks, whereas the **Team Member** can only view or update assigned tasks. By integrating **Users**, **Groups**, **Roles**, **Tables**, **ACLs**, and **Flow Designer**, the project establishes a seamless and transparent workflow that enhances accountability and collaboration.

During implementation, ServiceNow modules were configured to define users and groups, create custom roles, design Project and Task tables, assign permissions, and develop a workflow using Flow Designer. The workflow automatically updates task statuses, sends approval requests, and enforces access restrictions according to predefined rules. This automation significantly reduces manual intervention and potential human errors. Testing was carried out through user impersonation to verify that each role could perform only the intended operations.

The outcome of the project demonstrates improved security, efficiency, and clarity in access management. Users can now perform actions according to their designated roles, ensuring a controlled, transparent, and productive work environment. The project highlights how ServiceNow can be effectively utilized for **role-based access control (RBAC)** and **workflow automation**, ultimately optimizing team collaboration and governance. In the future, the system can be enhanced through integration with reporting dashboards, audit tracking, and automated alerts for policy violations to further strengthen operational intelligence and compliance.

1. Introduction

In organizations, poorly managed user permissions and unclear role assignments can lead to confusion, delays, and security risks. Without defined access boundaries, users may perform unauthorized actions or face difficulties collaborating effectively. To overcome these challenges, the “Optimizing User, Group, and Role Management” project was developed using **ServiceNow**, a powerful IT Service Management (ITSM) platform known for its flexibility and automation capabilities.

This project introduces a structured access control system by defining users (Project Manager and Team Member), assigning them to specific groups, and controlling their permissions through ACLs. Additionally, it incorporates an automated workflow to streamline task approvals and progress tracking. The primary objective is to build a secure, transparent, and efficient environment for user and role management.

In this project, the ServiceNow platform is utilized to demonstrate how automation and access control can simplify administrative processes. By leveraging ServiceNow’s modules - such as *User Administration, Groups, Roles, Access Control Lists (ACLs),* and *Flow Designer* - the system ensures that only authorized users can access relevant data and perform actions based on their assigned responsibilities. This structured approach minimizes manual intervention, reduces administrative overhead, and enhances the overall transparency of operations within the organization.

Furthermore, the implementation of automated workflows provides real-time tracking and approval mechanisms, eliminating delays and communication gaps between teams. The system not only strengthens security but also improves collaboration and accountability by clearly defining user boundaries and approval hierarchies. Overall, this project showcases how an efficient combination of *access control and workflow automation* in ServiceNow can lead to a more secure, organized, and productive IT environment, aligning with the modern standards of enterprise service management.

2. Problem Statement

Before implementing this solution, the organization faced several issues due to manual handling of access and role assignments. There was no proper distinction between user permissions, resulting in data modification risks and miscommunication. Team members lacked visibility into their assigned responsibilities, and project managers found it difficult to track task completion efficiently. The absence of structured workflows caused delays and accountability gaps.

Therefore, the project aims to develop a **role-based access management system** in ServiceNow that standardizes user responsibilities, applies proper access control mechanisms, and automates the approval workflow - ensuring both security and productivity.

3. Methodology / System Design

i. Design Approach

The solution was developed using ServiceNow Studio, focusing on simplicity, clarity, and access security. The project defines users, groups, and roles, each with specific permissions. Access Control Lists (ACLs) are applied to restrict data modification based on user roles. The Flow Designer module is used to automate workflow approvals and task status updates, ensuring a seamless and efficient process.

ii. System Architecture

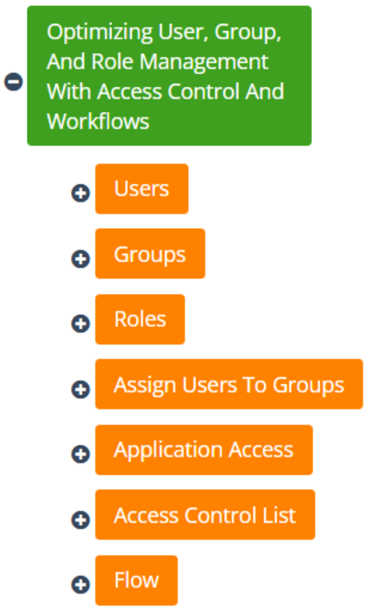
The architecture integrates the following ServiceNow components:

- **Users:** Project Manager (Alice) and Team Member (Bob)
- **Groups:** Project Team Group
- **Roles:** Project Manager Role, Team Member Role
- **Tables:** Project Table and Task Table
- **ACLs:** Role-based access permissions
- **Flow Designer:** Automated task approval and update workflow

Each component works together to form a secure structure. Alice (Project Manager) can create and assign tasks, while Bob (Team Member) can only view and update tasks assigned to him. Workflow automation ensures approvals and task status updates occur seamlessly.

iii. User Interface (UI) and User Experience (UX)

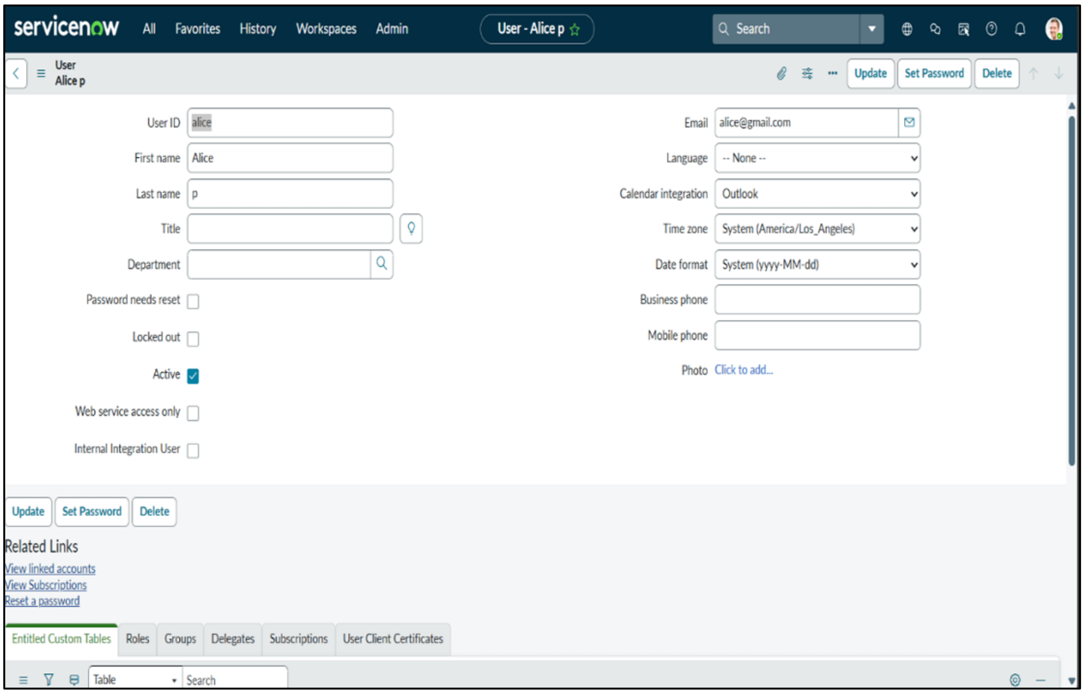
The interface is designed to be simple and consistent. The Project and Task tables are visible in the application menu, with form fields for project details, comments, and task status. Role-based visibility ensures that users only see data relevant to their permissions. A clean and structured layout enhances usability and workflow tracking.

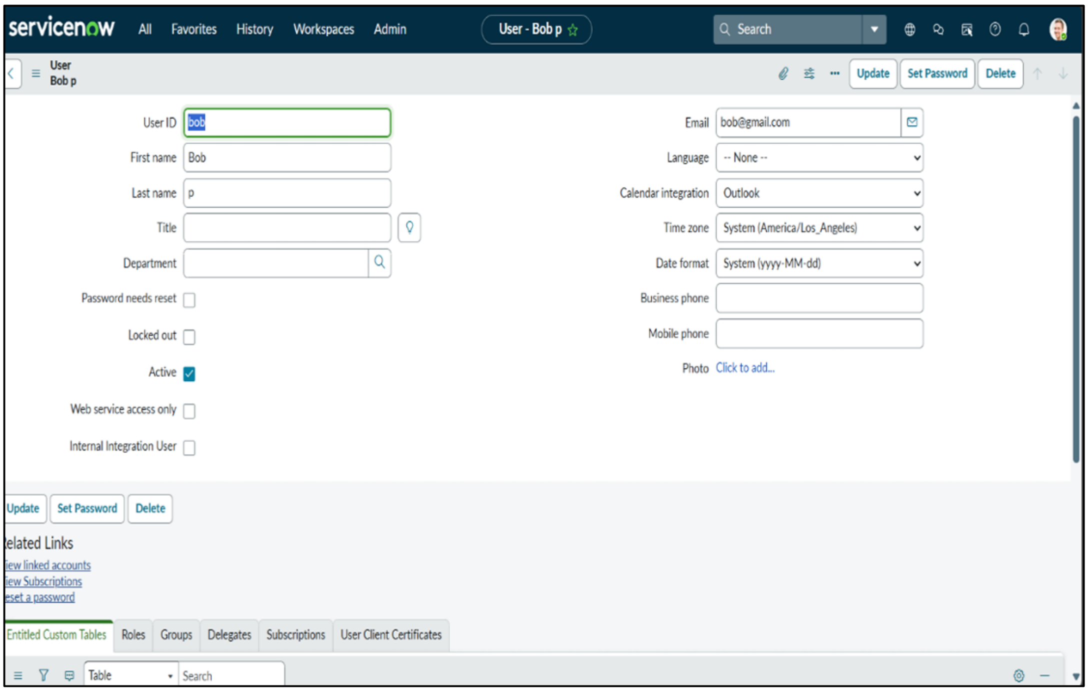


4. Implementation Details

STEP 1: Create Users

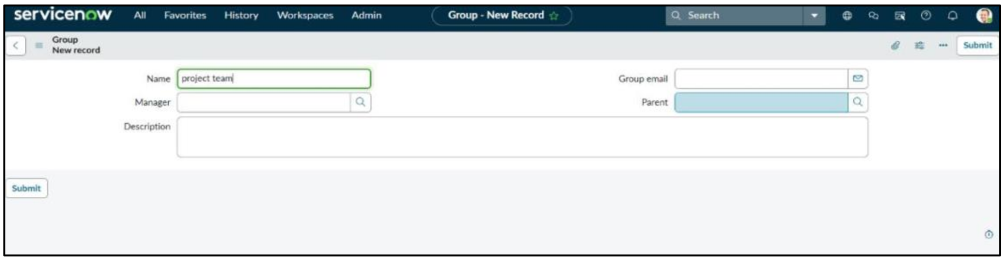
1. Navigate to **All → Users** under *System Security*.
2. Create two users:
    - Alice (Project Manager)
    - Bob (Team Member)
3. Click **Submit** to save both users.





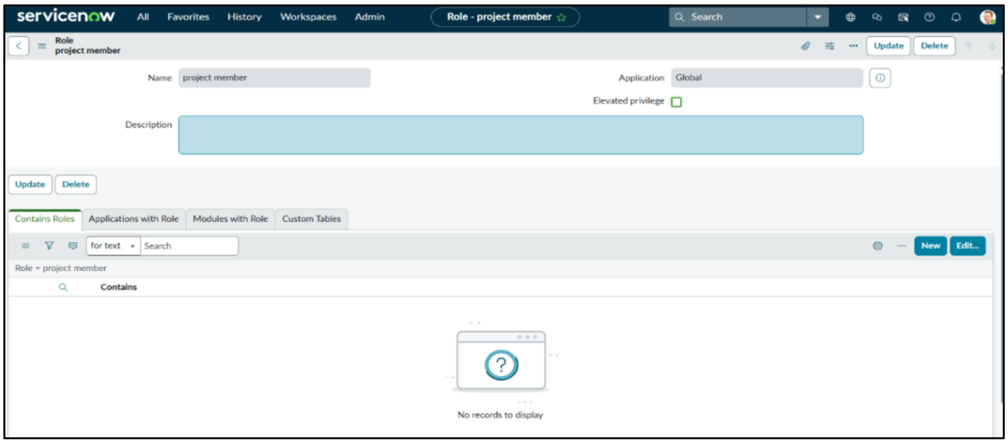
STEP 2: Create Groups

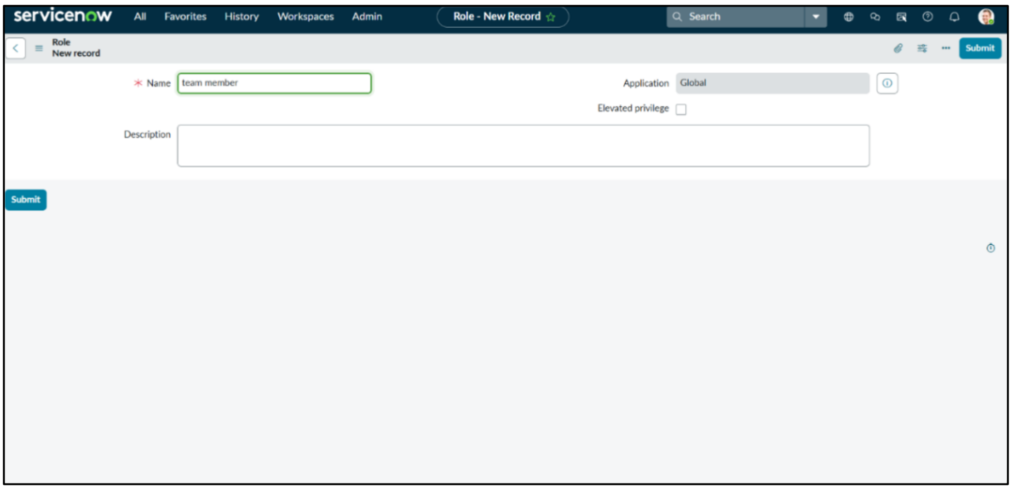
1. Go to **All → Groups**.
2. Create a new group named **Project Team**.
3. Add relevant details and click **Submit**.



STEP 3: Create Roles

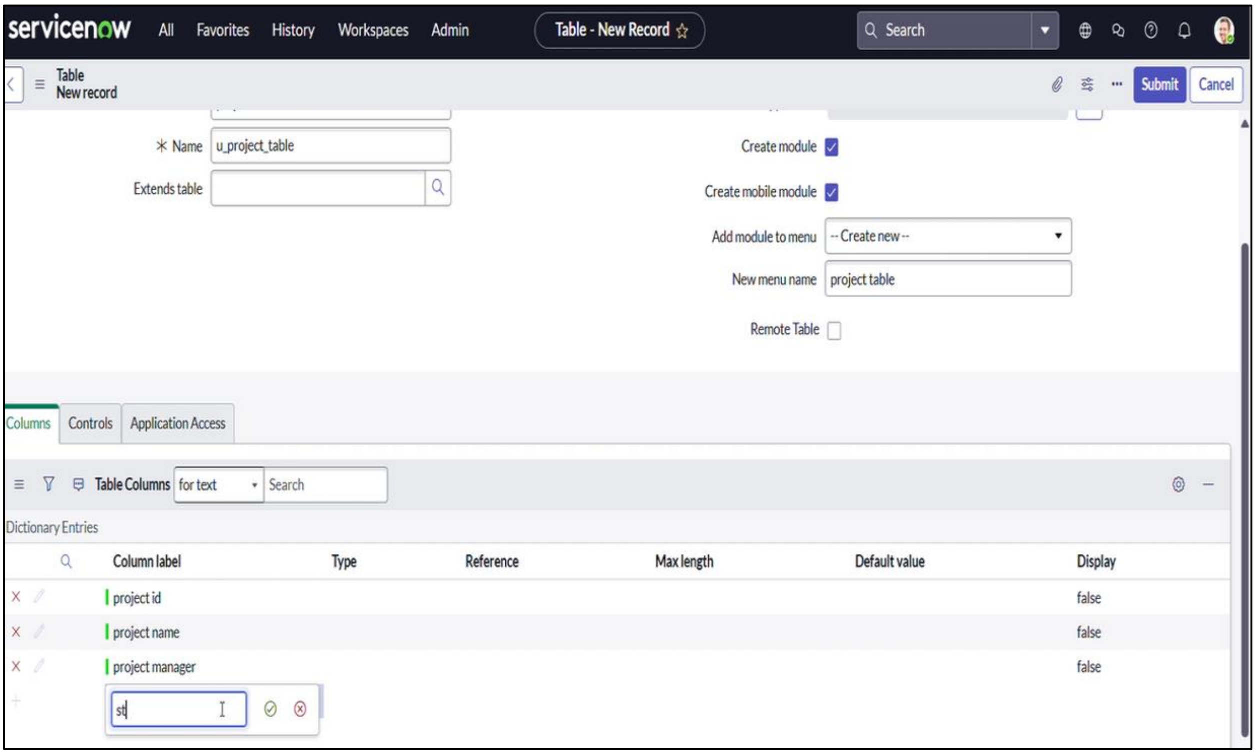
1. Navigate to **All → Roles**.
2. Create two roles:
   - **Project Manager Role**
   - **Team Member Role**
3. Click **Submit** after each role creation.

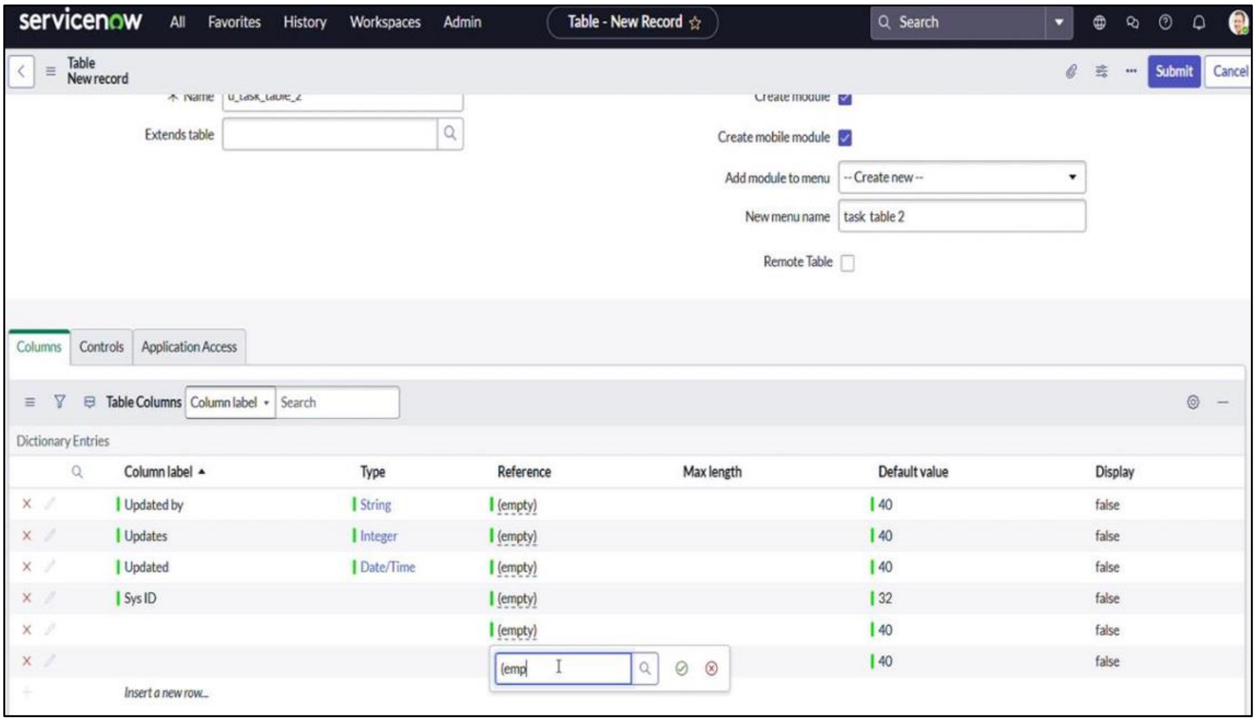


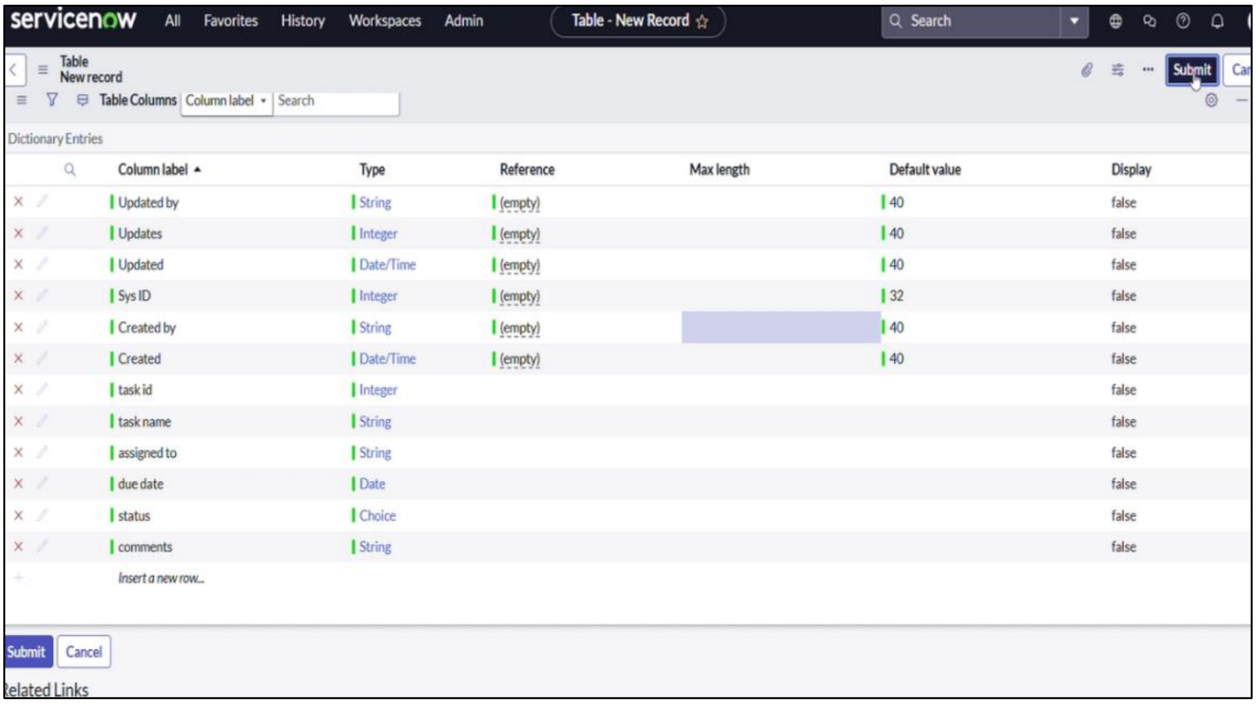


STEP 4: Create Tables

1. Go to **All → Tables** under *System Definition*.
2. Create a **Project Table** and enable *Create Module* and *Create Mobile Module*.
3. Add columns such as *Project Name*, *Assigned To*, *Status*, *Comments*, etc.
4. Similarly, create a **Task Table** with relevant fields.

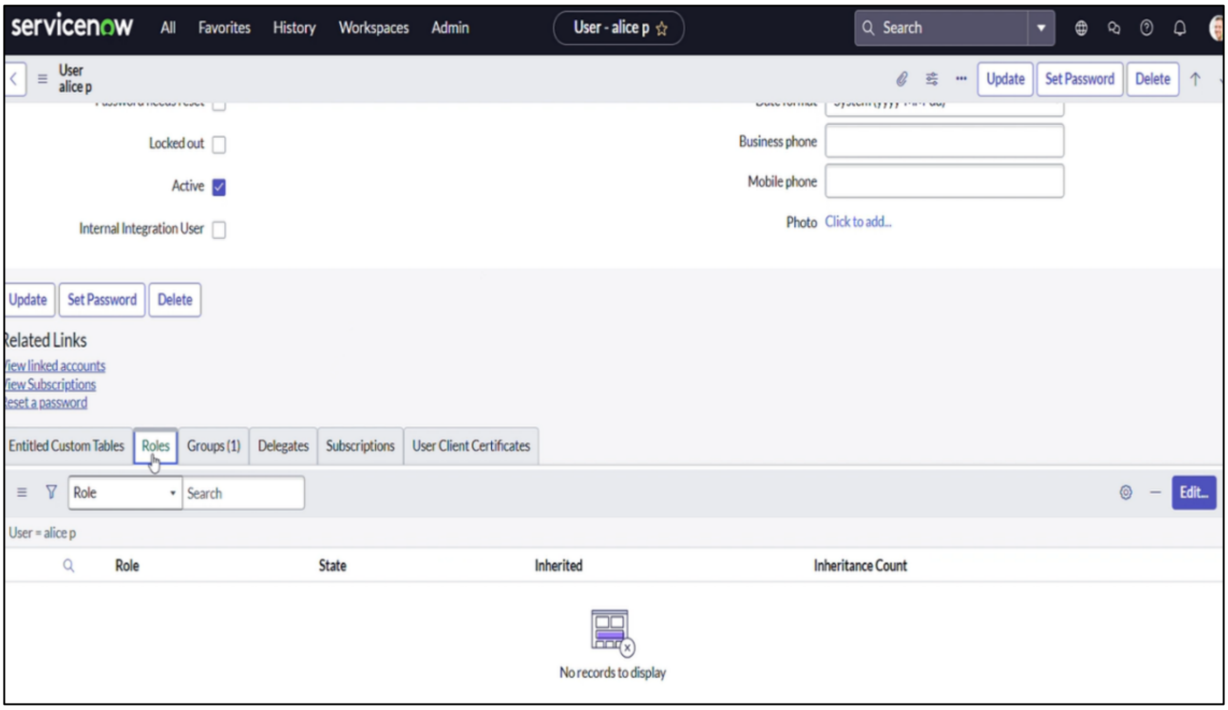






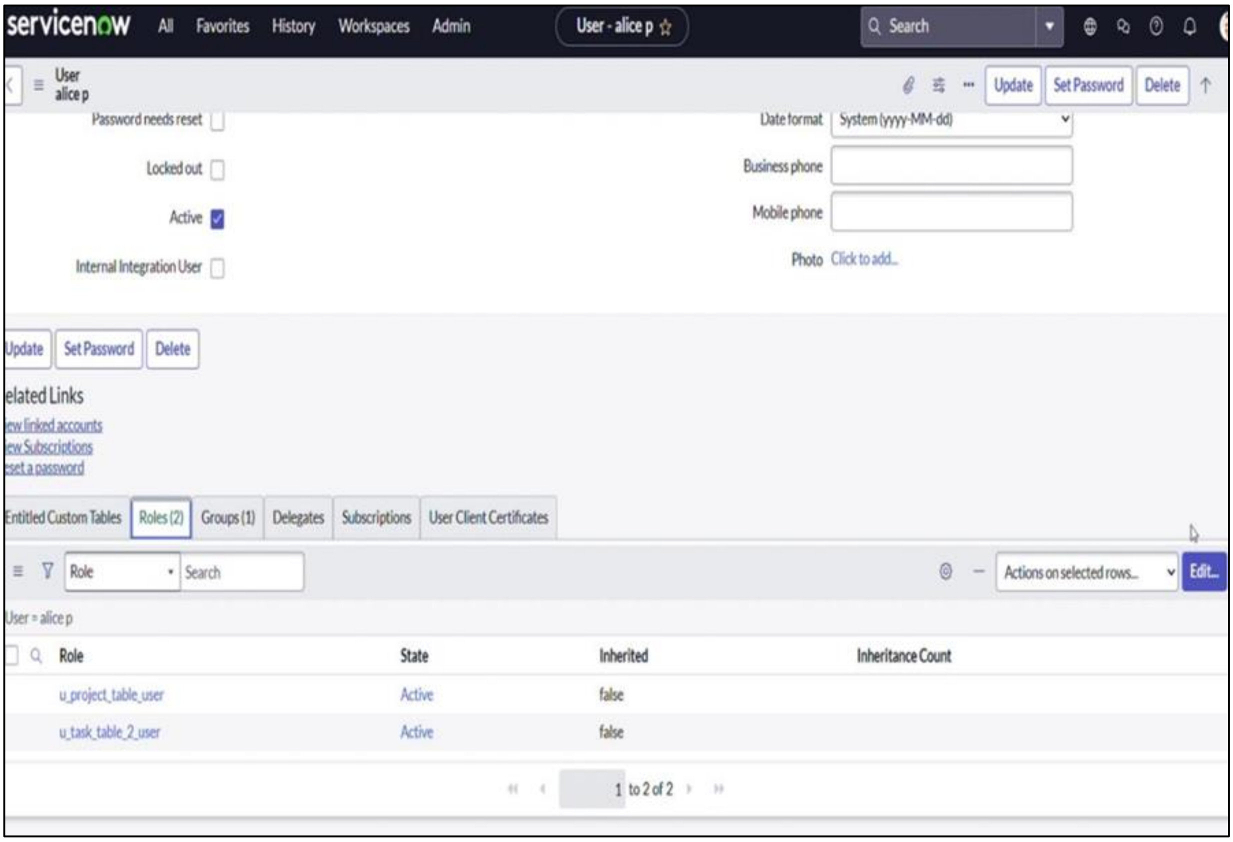
STEP 5: Assign Users to Groups

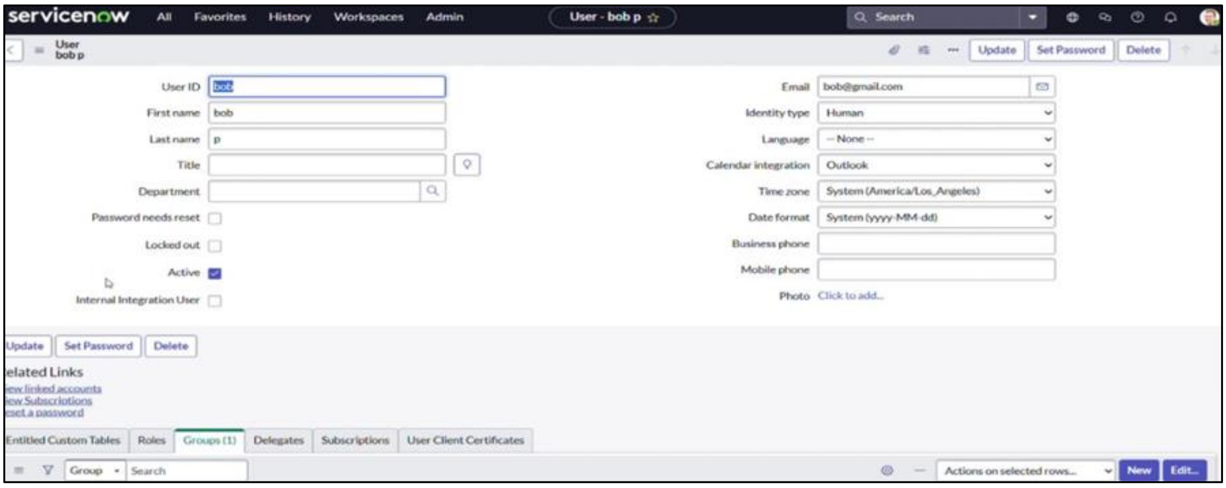
1. Open the **Project Team Group**.
2. Under *Group Members*, click **Edit**.
3. Add Alice and Bob to the group.
4. Save the changes.



STEP 6: Assign Roles to Users

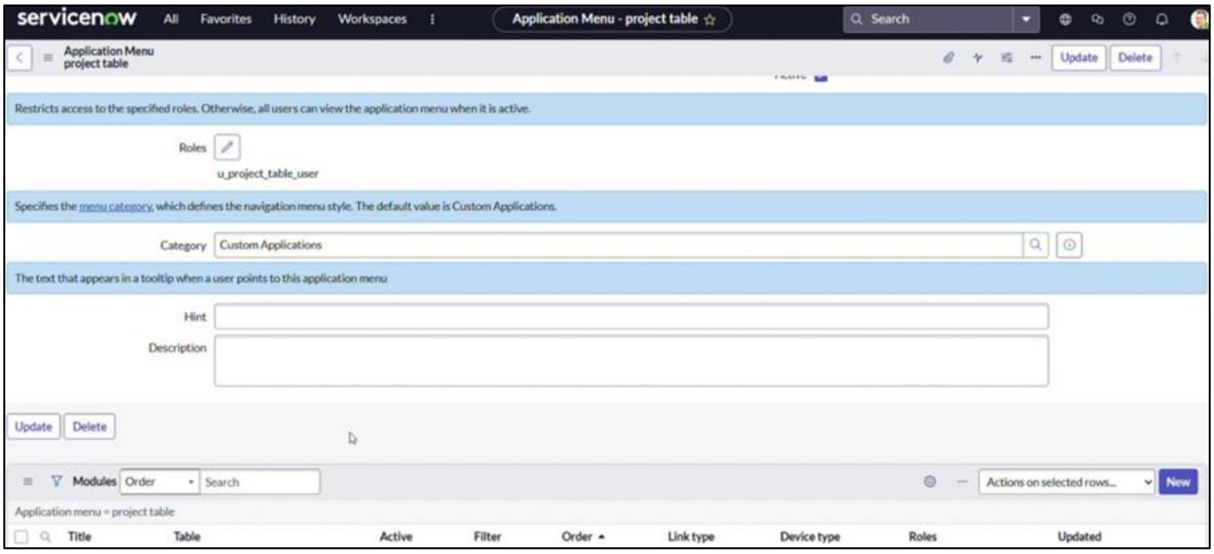
1. Assign **Project Manager Role** and **Table Roles** to Alice.
2. Assign **Team Member Role** and limited **Task Table Role** to Bob.
3. Use the *Impersonate User* option to verify role-based access.





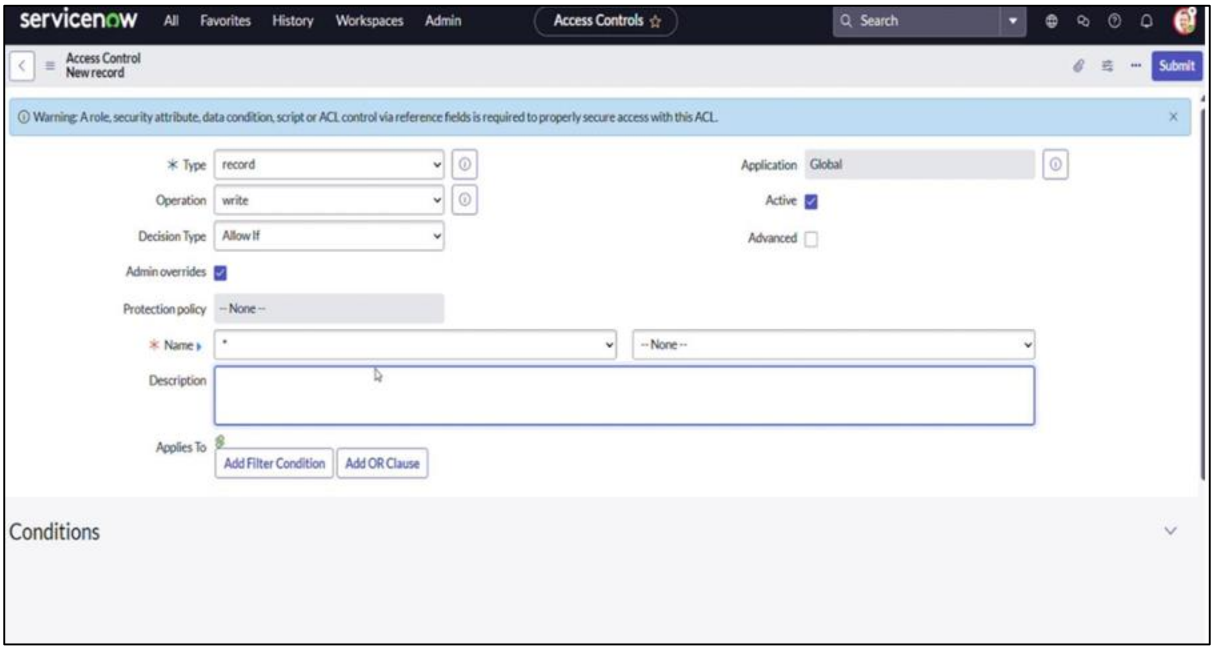
STEP 7: Application Access

1. Define which applications or tables each role can access.
2. Example: Project Manager → Full access to both tables; Team Member → Task Table only.



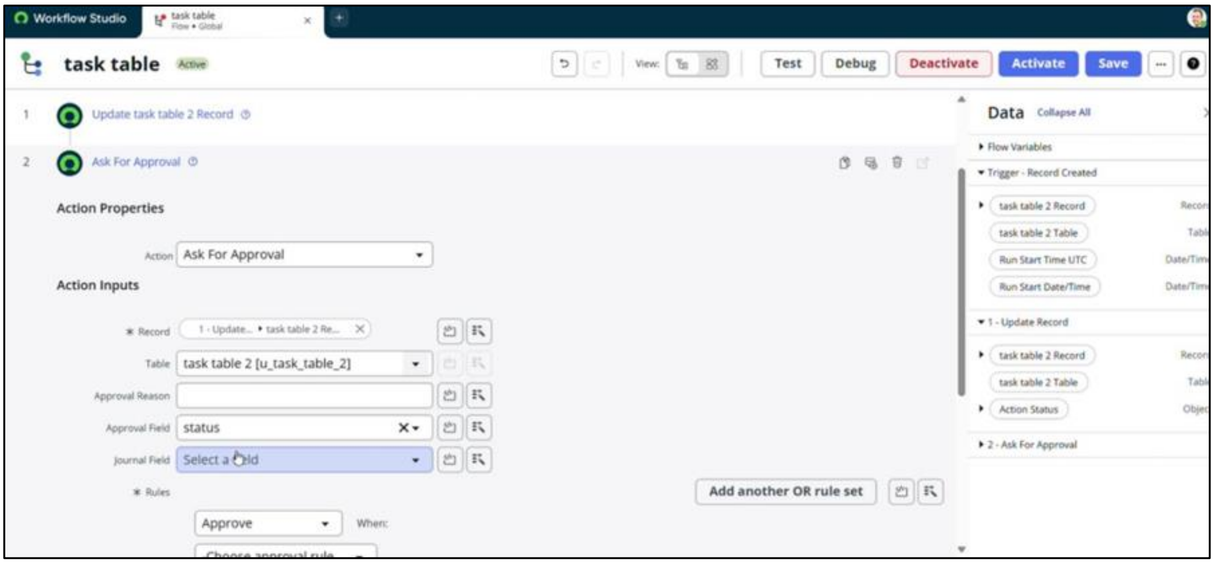
STEP 8: Access Control List (ACL)

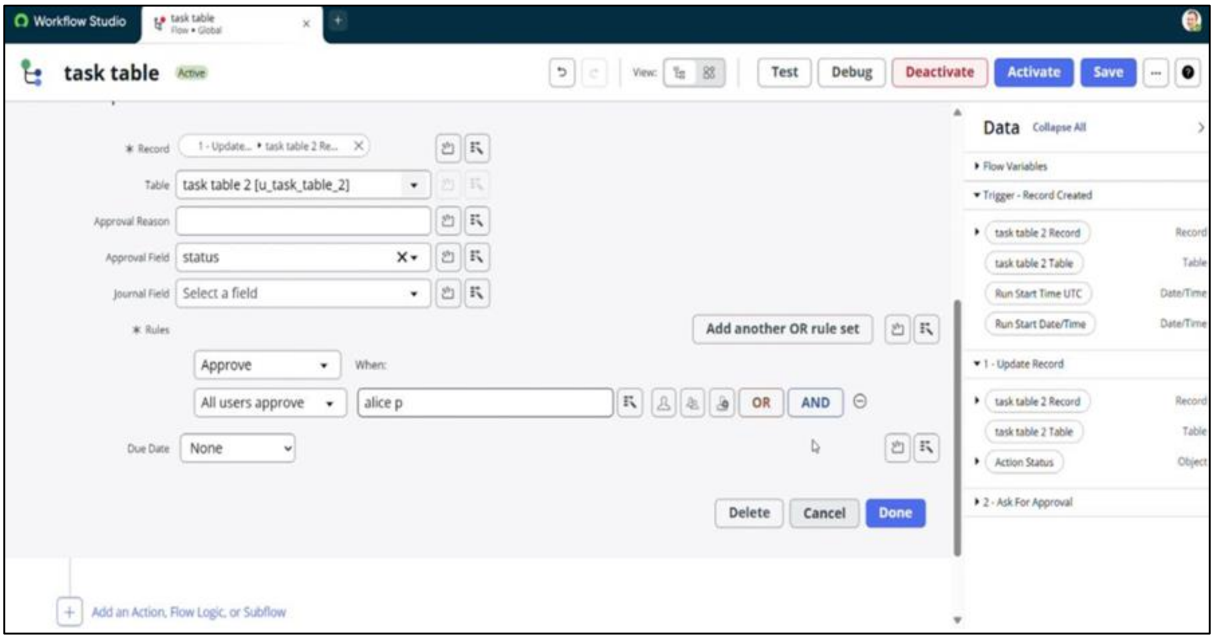
1. Create ACL rules under *System Security → Access Control (ACL)*.
2. Restrict access to sensitive fields.
   - Example: Bob can edit *Comments* and *Status* only.
3. Test access by impersonating users to verify restrictions.



STEP 9: Workflow Implementation (Flow Designer)

1. Open **Flow Designer → New Flow**.
2. Name the flow "Task Table Automation."
3. Set Trigger: *Record Created or Updated* on the Task Table.
4. Add Conditions:
   - Status = "In Progress"
   - Assigned to = Bob
5. Add Actions:
   - **Update Record:** Change status to "Completed."
   - **Ask for Approval:** Send approval to Alice.
6. Test the flow to ensure tasks are automatically updated and approved.

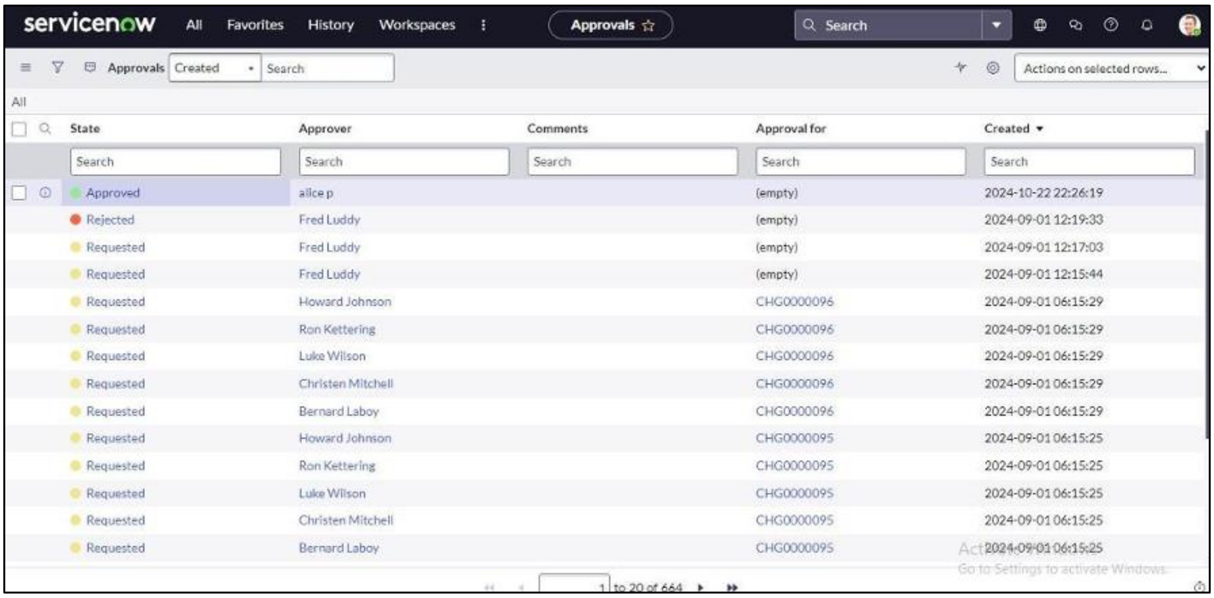




5. Testing

Testing was performed by impersonating both users:

- As **Bob (Team Member):** Could view and update assigned tasks only.
- As **Alice (Project Manager):** Could approve, monitor, and complete project tasks. The access control and workflow automation functioned correctly, ensuring secure and role-specific operations.



6. Future Scope

The project **“Optimizing User, Group, and Role Management through Access Control and Workflows”** successfully implemented a structured role-based access control system and automated workflow management using the ServiceNow platform. While the present version efficiently manages users, groups, and roles with proper access control and task automation, there is significant scope for enhancement and scalability in future implementations.

One of the key areas for future development is the **integration of analytics and reporting modules** within the ServiceNow environment. This would allow administrators to track user performance, access patterns, and workflow efficiency through visual dashboards. Automated reports can help management analyze bottlenecks, monitor system usage, and make informed decisions to optimize team performance.

Another potential improvement lies in introducing **multi-level and dynamic approval workflows**. In larger organizations, projects often require approvals from multiple departments or hierarchies. By integrating adaptive workflows that automatically route requests based on user roles or departments, the system can handle complex approval structures more effectively and reduce manual oversight.

Additionally, **integration with external systems** such as email clients, inventory databases, or HR management tools can make the system more comprehensive. For instance, integrating with HR systems would allow automatic creation or deactivation of users based on employment status, ensuring consistent data management and security.

The implementation can also be extended to include **real-time notifications and AI-based access recommendations**. With machine learning models, ServiceNow can predict access requirements, detect anomalies in user activity, and suggest optimal role configurations, thus enhancing security and operational intelligence.

In the long term, the project can evolve into a **centralized access management solution** across various enterprise platforms, ensuring compliance, transparency, and efficiency at all levels. These future enhancements would make the system more robust, intelligent, and adaptable to the growing demands of digital enterprises, aligning with the vision of creating a smarter, automated, and secure workflow environment.

7. Conclusion

The project **"Optimizing User, Group, and Role Management through Access Control and Workflows"** successfully achieved its objective of establishing a systematic and secure access management system using the ServiceNow platform. Through the effective use of ServiceNow tools such as *Users, Groups, Roles, Tables, Access Control Lists (ACLs),* and *Flow Designer*, the project streamlined administrative operations and improved control over user permissions within an organization.

The developed system ensures that every user receives the right level of access based on their assigned role and responsibilities. By automating approval workflows and integrating role-based access control mechanisms, the project minimizes manual errors, enhances transparency, and ensures accountability in task execution. This approach not only saves time but also strengthens data security by preventing unauthorized access and simplifying permission audits.

Furthermore, the project demonstrated how ServiceNow's low-code/no-code capabilities can be leveraged to design efficient workflows that improve operational productivity. The testing phase confirmed that users, groups, and roles were managed accurately and that the automated workflows performed as expected, providing real-time updates and notifications at every stage.

Overall, this project serves as a practical example of how automation can transform administrative management into a more organized and reliable process. It highlights the importance of integrating technology-driven solutions in organizational access management and lays a strong foundation for future developments in workflow optimization and enterprise automation.

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